Customer Revenue & Retention Analysis

Project report by Emmanuel Adebomi (Emmizee) | Tools: Python (pandas) in Jupyter Lab, Power BI Desktop | Data: Sales & Marketing Customer Dataset (Kaggle, synthetic)

1. Overview

Business context. I analysed the customer records of a subscription-style online business: 15,000 customers who signed up between January 2022 and September 2024, with their spend, acquisition channel, satisfaction, support activity and churn status. I approached it the way I would approach a request from a head of growth, starting from a decision rather than from a chart.

The business decision. Should next quarter's growth plan focus on acquiring new customers or on reducing churn among existing ones?

Questions I set out to answer

1. Is the business growing, and is that growth coming from more customers or from higher spend? (Finance)
2. Which customer segments carry the revenue, and are the differences between them real? (Finance)
3. Is average order value stuck, and where could it be lifted? (Finance)
4. Is acquisition spend efficient, and does any channel deserve more or less budget? (Marketing)
5. Where is revenue leaking through refunds and discounts? (Finance and Operations)
6. Which signal predicts customer loyalty: NPS or satisfaction? (Customer Success)
7. Who is leaving, and what do they have in common? (Customer Success)
8. How much revenue is tied to customers at risk? (Customer Success and Finance)
9. Do acquisition channel or signup timing affect retention? (Marketing and Product)

North Star metrics. I anchored the analysis on three metrics, defined in plain terms:

- **Churn rate:** the share of customers who have left (15.3%).
- **CAC:** the average marketing spend per customer (17.58).
- **ARPU:** average revenue per customer with recorded spend (524.36).

Scope and honesty note. The dataset is synthetic, so my findings demonstrate method and judgement, not the performance of a real company. It also has no cost of goods, no order dates and no visitor-to-order funnel. That means I could not compute a true gross margin, a monthly sales trend or a visit-to-order conversion rate. Where I needed a substitute, I used a clearly labelled proxy instead of inventing numbers.

2. Executive summary

Between January 2022 and September 2024, the business acquired 15,000 customers (13,950 with recorded spend) and generated 7.31M in revenue, at an ARPU of 524 and an AOV of 60. The business is stable but flat: revenue (about 220k a month), new customers (about 455 a month), AOV and ARPU are almost identical in all three years, and no acquisition channel stands out on cost, return or retention. The clearest lever is retention. 15.3% of customers (2,298) have churned, and every one of them showed at least one of three warning signs: a satisfaction score of 1 or 2, five or more support tickets, or total spend under 300. Customers with none of these signs show 0.0% churn, while the flagged 38% of customers hold about a quarter of total revenue.

Answer to the decision: focus next quarter on retention. Acquisition is cheap and flat, with no channel to scale, while churn is concentrated in customers who can be identified in advance.

3. The data

Source. The Sales and Marketing Dataset on Kaggle: one table of 15,000 customers and 30 columns. I chose it because it was the only candidate I found where acquisition cost, NPS, lifetime value and churn exist as native fields, so I did not have to assume any of them. No public dataset I could find has real customer-level ad spend, so synthetic data was the trade-off for that.

Structure. Because the data is a single table, there are no joins and an entity-relationship diagram would add nothing. Instead, I grouped the columns by business area:

Group	Columns
Customer profile	customer_id, gender, age, country, city, device_type
Time	signup_date, last_purchase_date

Group	Columns
Acquisition	acquisition_channel, marketing_spend_per_user
Plan	subscription_type, is_premium_user
Engagement	total_visits, avg_session_time, pages_per_session, email_open_rate, email_click_rate
Money	total_spent, avg_order_value, discount_used, coupon_code, payment_method, lifetime_value
Service and loyalty	support_tickets, refund_requested, delivery_delay_days, satisfaction_score, nps_score, last_3_month_purchase_freq, churn

What the data can and cannot answer. It can answer questions about revenue per customer, acquisition cost, customer satisfaction and churn. It cannot answer monthly sales trends (total_spent is a lifetime total with no order dates), gross margin (no product cost) or visit-to-order conversion (no funnel events).

4. Data preparation and cleaning

4.1 My principles

1. **Never invent a value.** If a number was missing, I labelled it or left it empty instead of filling it with an assumption.
2. **Flag, do not repair.** Where two columns contradicted each other, I added a flag column instead of choosing which one was right.
3. **Keep every customer.** I deleted no rows, so all 15,000 customers remain in the clean file.
4. **Test before trusting.** I checked each assumption against the data before building on it.

4.2 Inspection

I loaded the CSV into pandas and ran `info()` and `describe()`. This confirmed 15,000 rows and 30 columns, five columns with missing values, and both date columns stored as text. I then read the first 20 rows closely, which exposed problems that summary statistics hid: dates in the wrong order, coupon codes that disagreed with the discount flag, and cities that did not belong to the listed country.

4.3 Issue log

#	Issue	Evidence	What I did	Why
1	Dates stored as text	signup_date and last_purchase_date had type object	Converted both with `pd.to_datetime(..., errors='coerce')`	Needed for time axes and date logic
2	Missing gender	738 rows (4.9%)	Filled with "Unknown"	Keeps every customer and invents nothing
3	Missing age	1,200 rows (8.0%)	Left empty and grouped into an "Unknown" age band	A median fill would distort segment results
4	Missing satisfaction score	702 rows (4.7%)	Left empty and excluded from satisfaction charts	A blank is not a zero, and averages skip it
5	Missing total_spent	1,050 rows (7.0%)	Left empty, added a `spend_recorded` flag, and calculated revenue KPIs on the 13,950 customers who have a value	Their churn rate was 10.0% versus 15.7% for the rest, so the gap was not random. A mean fill would hide that and dropping the rows would bias the results

#	Issue	Evidence	What I did	Why
6	Missing coupon_code	6,133 rows (40.9%)	Left as it was	I first assumed blank meant no coupon, then tested it. 4,537 customers had a code but `discount_used` of 0, and 3,087 had no code but `discount_used` of 1, so 7,624 rows (51%) contradicted each other. Filling "No Coupon" would have baked in a false assumption
7	Last purchase before signup	3,762 rows (25.1%)	Added a `date_valid` flag. I did not swap or repair dates	I cannot know which date is wrong. I used signup_date as the time axis and the churn flag for retention, which depend on neither
8	Lifetime value below recorded spend	2,340 rows (about 17% of customers with spend)	Kept LTV as provided and labelled it that way on the dashboard	Too widespread to ignore, and not mine to correct. I did not use it as a headline

#	Issue	Evidence	What I did	Why
9	City does not match country	Obvious in the first 20 rows (for example Berlin listed under India), and payment methods did not match countries either	Used country at most and made no geographic claims	A geographic story would rest on fields I could not trust

4.4 Fields I created

- **spend_recorded:** true when total_spent has a value, so revenue KPIs can state their coverage.
- **date_valid:** true when last_purchase_date is on or after signup_date.
- **signup_month:** the month of signup, used as the time axis for cohort charts.
- **age_band** and **age_band_order:** Youth (under 25), Young Adults (25-34), Adults (35-54), Senior Adults (55+) and Unknown, with an order column so the bands sort correctly in Power BI.
- **In Power BI:** calculated columns for spend bands and for risk flags (one point each for satisfaction of 1-2, five or more support tickets, and spend under 300), with blanks handled explicitly so customers who did not answer a question were not flagged by accident.

4.5 Checking whether the data had signal

Before building dashboards, I tested whether the columns related to each other, because a synthetic dataset can be random noise. Average churn, NPS, spend and acquisition cost were almost identical across all five channels. Correlation with churn was near zero for almost every field, with three exceptions: satisfaction score (-0.30), total spend (-0.16) and support tickets (+0.13). That told me where the real story was, and that channel comparisons would be flat.

4.6 Validation

I calculated every headline KPI in pandas first and then confirmed the Power BI measures matched:

Metric	Value
Customers	15,000
Total revenue	7,314,785.69
ARPU (13,950 customers with spend)	524.36
AOV	60.08
CAC (marketing spend per customer)	17.58
Churn rate	15.3%
NPS	-46.2

4.7 What I deliberately did not do

- I did not impute any money field.
- I did not delete any customer.
- I did not repair dates.
- I did not invent a cost of goods, an order count or a visit funnel.

5. Dashboard design

I built four Power BI pages with the same layout: page navigation along the top, slicers on the left, KPI cards, trend and breakdown charts, and an insights and recommendation box. Each page is tied to a question a stakeholder would ask.

KPI	Definition
Total revenue	Sum of total_spent for customers with a recorded value
ARPU	Total revenue divided by customers with recorded spend
AOV	Average of avg_order_value, a customer-level average
CAC	Average marketing_spend_per_user

KPI	Definition
Return on acquisition spend	Revenue divided by acquisition spend, about 30
Churn and retention	Share of customers with churn = 1, and one minus that
Premium conversion	Share of customers on a premium plan, 30.4% (used in place of visit-to-order conversion)
NPS	Percentage of promoters (9-10) minus detractors (0-6)
LTV	Average lifetime_value, shown as provided

Proxies I used instead of missing metrics. Revenue is shown by signup cohort, not by month of sale. Gross margin is replaced by return on acquisition spend. I did not show month-over-month change on revenue cards, because the data has no dated sales.

6. Insights deep dive

6.1 Money In

Insight 1: the business is stable, not growing.

Year	Revenue	AOV	ARPU
2022	2.67M	60.28	523.29
2023	2.67M	59.94	531.23
2024 (Jan-Sep)	1.97M	60.00	516.74

At first glance, 2024 looks like a 26% decline. It is not: the data stops in September, and per month, revenue is about 222k, 222k and 219k, with roughly 455 new customers a month throughout. *Why it matters:* growth will not appear unless something changes on purpose. *Action:* treat flat acquisition and flat AOV as the baseline to beat.

Insight 2: Senior Adults look valuable, but a few large spenders create that picture. Senior Adults have the highest ARPU (about 1,079 versus about 507 for Adults) and the lowest revenue (about 383k, roughly 5% of the total, from 355 customers). Checking the median changed the story:

Age band	Customers with spend	Mean	Median	Highest
Adults (35-54)	6,350	507.0	494.9	1,404.8
Senior Adults (55+)	355	1,079.0	539.8	15,910.4
Young Adults (25-34)	4,276	515.5	506.6	1,475.0
Youth (under 25)	1,853	515.5	499.5	1,489.3
Unknown	1,116	495.3	476.4	1,539.3

The typical senior spends only about 9% more than the typical adult. One senior spent about ten times more than the top spender in any other group, and capping the top 1% of spenders drops the senior average from 1,079 to 569, only about 12% above Adults. *Why it matters:* my first instinct was to recommend growing the senior segment, and the median showed that would have been built on a few accounts. *Action:* identify those accounts, and pilot before investing.

Insight 3: AOV has not moved in three years. It sits at about 60 every year. I have no product data, so lifting it is a hypothesis to test (bundles, a free-delivery threshold set slightly above 60), not a finding.

Insight 4: payment, channel and country rankings are noise. The top payment method was UPI in 2022, PayPal in 2023 and BKash in 2024, with shares between about 18.6% and 21.4%. The top channel moved from Organic to Google Ads, and the top country from Germany to Bangladesh. Because the leaders change every year and the gaps are small, I do not recommend acting on any of them. *Action:* none, which is itself a finding.

6.2 Money Out

Insight 5: no channel earns a bigger budget. CAC is about 17.58 overall and ranges only from 17.24 to 17.81 across channels, including Organic, which is normally near free and is another sign the data is synthetic. Each 1 of acquisition spend is matched by about 30 in customer revenue in every channel. *Action:* hold channel budgets, and track CAC over time to see whether it rises as volume grows.

Insight 6: refunds and discounts are the real leak. 15% of customers requested a refund and about half (49.4%) used a discount. Refund requests have almost no link with churn (correlation -0.007), so they look like a separate problem from retention. *Action:* find which plans or channels drive refunds, and test whether discount-led sales are the cause. I have not yet run that breakdown.

6.3 Customer Behavior

Insight 7: NPS tells nothing here, and satisfaction tells a lot. NPS is -46.2, but scores are spread almost perfectly evenly across 0 to 10 (between 1,303 and 1,417 customers at each score), which mechanically produces a very negative NPS. It has no relationship with satisfaction (-0.009) or churn. Satisfaction, with an average of 3.59 out of 5, does:

Satisfaction	Customers	Churn
1	753	48.7%
2	1,455	50.2%
3	3,739	9.7%
4	5,323	8.9%
5	3,028	9.7%

Why it matters: a metric that does not predict loyalty cannot guide retention. *Action:* use satisfaction to flag at-risk customers, and validate how NPS is collected before relying on it.

Insight 8: premium status does not change retention. 30.4% of customers are on premium plans, and premium status has almost no correlation with churn (-0.005). Upgrading customers is therefore not, on this evidence, a retention tool.

6.4 Funnel Health

Insight 9: churn is concentrated, not random. Three signals separate churners from the rest:

Signal	Churn when present	Churn otherwise
Satisfaction of 1-2	about 50%	about 9-10%
5 or more support tickets (809 customers)	50.4%	12-14%
Spend under 300	about 34%	about 9-10%

Churn stays flat at 12-14% from zero to four tickets and jumps only at five, so five tickets is the trigger point, not three or four.

Insight 10: the warning signs account for every churned customer.

Risk flags	Customers	Revenue	Churn
0	9,294	5,481,432	0.0%
1	4,939	1,672,334	38.7%
2	743	157,291	50.2%
3	24	3,729	54.2%

All 2,298 churned customers sit in the flagged group of 5,706 customers (38%), which holds 1.83M, about a quarter of total revenue. *Why it matters:* the customers most likely to leave can be listed today, before they leave. A group with exactly 0.0% churn would never occur in a real business, which tells me the data was generated from rules. The method still transfers to real data. *Action:* build an early-warning list and follow up on these customers first.

Insight 11: where customers came from, and when, does not predict whether they stay. Churn is between 14.8% and 15.9% in every channel. That closes off a tempting but unsupported conclusion: reallocating budget between channels will not reduce churn.

7. Recommendations

Priority	Action	Evidence	Owner	Metric to track
P0	Launch an early-warning retention programme for flagged customers	Churn is 38.7%, 50.2% and 54.2% at 1, 2 and 3 flags and 0.0% at none	Customer Success	Churn among flagged customers
P1	Escalate at five support tickets and follow up on satisfaction of 1-2 within days	Churn is 50.4% at 5+ tickets and about 50% at satisfaction 1-2	Support	Churn at 5+ tickets
P1	Hold channel budgets and do not reallocate on current evidence	Cost, return and churn differ by under 3% across channels	Marketing	Return per 1 spent and CAC by channel

Priority	Action	Evidence	Owner	Metric to track
P2	Investigate the large senior accounts and any revenue concentration	The senior mean is 1,079 against a median of 540	Finance	Revenue share of the largest accounts
P2	Investigate refund drivers and the role of discounts	15% refund rate, 49.4% discount use	Operations	Refund rate by plan and channel
P3	Use satisfaction, not NPS, to track loyalty until NPS collection is validated	NPS is evenly spread and unrelated to churn	Customer Success	Satisfaction versus churn

I have left targets blank on purpose. Setting them responsibly needs a pilot, not a guess.

8. Caveats and assumptions

- **The data is synthetic.** Findings show the method, not the behaviour of a real company, and several patterns (0.0% churn in the unflagged group, near-identical channels, evenly spread NPS) are signs of generated data.
- **Revenue is lifetime, not monthly.** Time charts show revenue by signup cohort. 2024 contains nine months only.
- **Revenue KPIs cover 13,950 customers.** 1,050 customers have no recorded spend, so true revenue is probably higher.
- **Gross margin is not available**, because there is no product cost, and conversion is limited to premium conversion.
- **Lifetime value is shown as provided** and contradicts recorded spend in about 17% of cases.
- **Dates, coupons and geography are inconsistent** and were flagged, not corrected.
- **Small groups need caution.** Senior Adults are 355 customers, and country-by-month patterns rest on very few customers.

9. What I would do next

1. Repeat the method on real business data, where the patterns will be messier and the conclusions more reliable.
2. Replicate the key calculations in SQL and link them alongside the notebook.
3. Break refunds down by plan, channel and discount use.
4. Identify the largest senior accounts and calculate revenue concentration.
5. If real order dates become available, add true monthly revenue, cohort retention and month-over-month change.